



1. Use your knowledge of the CAST Rule and radians to determine the **exact rationalized value** of the question below.

$$\frac{\sec \frac{5\pi}{3} - \tan \frac{11\pi}{6}}{\sin \left(-\frac{\pi}{4}\right)}$$

$$\frac{2 - \left(\frac{-1}{\sqrt{3}}\right)}{\frac{-1}{\sqrt{2}}}$$

(5)

	Diagram	Ratio
$\sec \frac{5\pi}{3}$ $\frac{1}{\cos \frac{5\pi}{3}} = \frac{h}{x}$		2 ✓
$\tan \frac{11\pi}{6}$		$\frac{-1}{\sqrt{3}}$ ✓
$\sin \left(-\frac{\pi}{4}\right)$		$\frac{-1}{\sqrt{2}}$ ✓

$$= \frac{\frac{2\sqrt{3}}{\sqrt{3}} + \frac{1}{\sqrt{3}}}{\frac{-1}{\sqrt{2}}}$$

$$= \frac{2\sqrt{3} + 1}{\sqrt{3} \left(\frac{-1}{\sqrt{2}}\right)}$$

$$= \frac{(2\sqrt{3} + 1)(-\sqrt{2})}{\sqrt{3}}$$

$$= \frac{-2\sqrt{6} - \sqrt{2}}{\sqrt{3}} = \frac{-6\sqrt{2} - \sqrt{6}}{3}$$

2. Determine the exact radian angle(s) for each of the following ratios where $x \in (0, 2\pi)$. Draw a diagram for each.

(4)

a) $\cos \theta = -\frac{1}{\sqrt{2}}$

$\pi - \frac{\pi}{4} = \frac{3\pi}{4}$
 $\pi + \frac{\pi}{4} = \frac{5\pi}{4}$

$\theta = \frac{3\pi}{4}, \frac{5\pi}{4}$

b) $\csc \theta = -\frac{2}{\sqrt{3}} = \frac{1}{\sin \theta}$

$\pi + \frac{\pi}{3} = \frac{4\pi}{3}$
 $2\pi - \frac{\pi}{3} = \frac{5\pi}{3}$

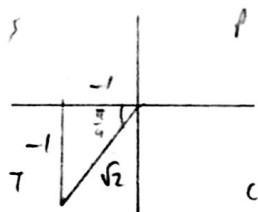
$\theta = \frac{4\pi}{3}, \frac{5\pi}{3}$

3. A small fan turns 2400 revolutions per hour. Express the angular velocity, in radians per second as an exact answer.

$$\frac{2400 \text{ rev}}{1 \text{ h}} \times \frac{1 \text{ h}}{60 \text{ min}} \times \frac{1 \text{ min}}{60 \text{ sec}} \times \frac{2\pi \text{ rad}}{1 \text{ rev}} = \frac{4\pi}{3} \text{ rad/sec}$$

(1)

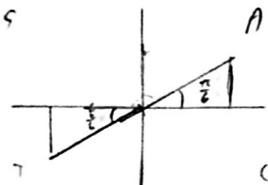
4. Determine the exact primary trig ratios for $\frac{5\pi}{4}$.



$$\sin \frac{5\pi}{4} = -\frac{1}{\sqrt{2}} \quad \cos \frac{5\pi}{4} = -\frac{1}{\sqrt{2}} \quad \tan \frac{5\pi}{4} = 1$$

(2)

5. Determine another value between $0 \leq x \leq 2\pi$ that is equivalent to $\tan \frac{7\pi}{6}$.



$$\tan \frac{\pi}{6}$$

(1)

6. A large pipe has a circumference of 120 meters, which is to be placed underground. Engineers need a portion of the pipe to connect to another pipe. Determine the maximum arc length of the pipe they could use to connect if the maximum central angle being used is $\frac{19\pi}{12}$.

$$\begin{aligned} \theta &= \frac{a}{r} & 2\pi r &= 120 & a &= \frac{19\pi}{12} \left(\frac{60}{\pi} \text{ m} \right) \\ r &= \frac{60}{\pi} & a &= 19(5) = 95 \text{ m} \end{aligned}$$

(2)

7. Matt is helping construct a triangular sandbox in the playground. Determine the area in exact simplified form.

$$\overline{CD} = \overline{DB} = \frac{\overline{CB}}{\sqrt{2}} = \frac{6\sqrt{2}}{\sqrt{2}} = 6 \text{ m}$$

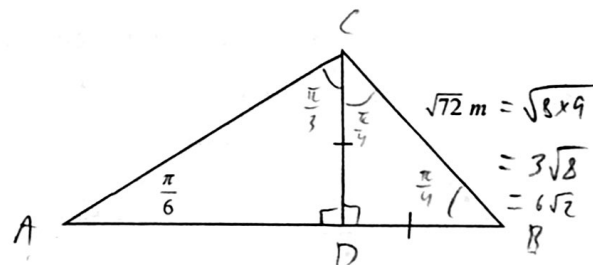
$$\begin{aligned} \overline{AD} &= \overline{CD} \times \sqrt{3} \\ &= 6\sqrt{3} \text{ m} \end{aligned}$$

$$\begin{aligned} \text{Area} &= \frac{1}{2}bh \\ &= \frac{1}{2}(\overline{AD} + \overline{DB})(\overline{CD}) \end{aligned}$$

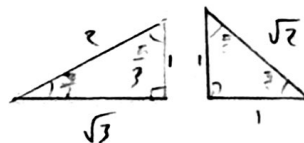
$$= \frac{1}{2}(6\sqrt{3} + 6)(6)$$

$$= 3(6\sqrt{3} + 6)$$

$$= (18 + 18\sqrt{3}) \text{ m}^2$$



(3)



8. Prove the following identities.

a) $(\sin x + \cos x) \left(\frac{\tan^2 x + 1}{\tan x} \right) = \frac{1}{\cos x} + \frac{1}{\sin x}$

$$\begin{aligned} RS &= \frac{1}{\cos x} + \frac{1}{\sin x} \\ &= \frac{\sin x + \cos x}{\sin x \cos x} \\ &= (\sin x + \cos x) \left(\frac{1}{\sin x \cos x} \right) \\ &= (\sin x + \cos x) \left(\frac{\sin^2 x + \cos^2 x}{\sin x \cos x} \right) \\ &= (\sin x + \cos x) \left(\frac{\frac{\sin^2 x + \cos^2 x}{\cos^2 x}}{\frac{\sin x \cos x}{\cos^2 x}} \right) \end{aligned}$$

$\Rightarrow = (\sin x + \cos x) \left(\frac{\tan^2 x + 1}{\tan x} \right)$
 $= LS$

(3)

b) $\frac{\sec x + \tan x}{\sec x - \tan x} = \frac{1 + 2\sin x + \sin^2 x}{\cos^2 x}$

$$LS = \frac{\sec x + \tan x}{\sec x - \tan x}$$

$$= \frac{\sec x (1 + \sin x)}{\sec x (1 - \sin x)}$$

$$= \frac{(1 + \sin x)(1 + \sin x)}{(1 - \sin x)(1 + \sin x)}$$

$$= \frac{1 + 2\sin x + \sin^2 x}{\cos^2 x}$$

$$= RS$$

(3)

$$c) \frac{1 - \cos x}{\cos x} = \frac{\tan x}{\sec x}$$

$$L S = \frac{1 - \cos x}{\cos x}$$

$$= \frac{1 - \cos x}{\cos x} \times \cot x$$

$$= \frac{1 - \cos x}{\cos x} \times \frac{\cos x}{\sin x}$$

$$= \frac{1 - \cos x}{\sin x}$$

$$= \frac{1 - \cos x}{\sin x} \times \frac{1 + \cos x}{1 + \cos x}$$

$$= \frac{1 - \cos^2 x}{\sin x (1 + \cos x)}$$

$$= \frac{\sin^2 x}{\sin x (1 + \cos x)}$$

$$= \frac{\sin x}{1 + \cos x}$$

$$= \frac{\sin x}{1 + \cos x} \times \frac{\sec x}{\sec x}$$

$$= \frac{\sin x \sec x}{1 + \cos x} \times \frac{1}{\sec x}$$

$$= \frac{\frac{\sin x}{\cos x}}{1 + \cos x}$$

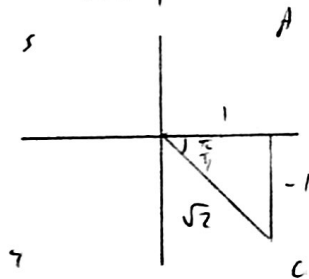
$$= \frac{\tan x}{1 + \cos x}$$

$$= R S$$

(3)

9. Simplify the following.

$$\sec \frac{7\pi}{4} = \sqrt{2}$$



$$\frac{5}{3} \sec \frac{7\pi}{4} - \frac{\csc \frac{5\pi}{4}}{\cos^2 \frac{\pi}{6}}$$

$$= \frac{5}{3} (\sqrt{2}) - \frac{(-\sqrt{2})}{\left(\frac{\sqrt{3}}{2}\right)^2}$$

$$= \frac{5\sqrt{2}}{3} + \sqrt{2} \left(\frac{4}{3}\right)$$

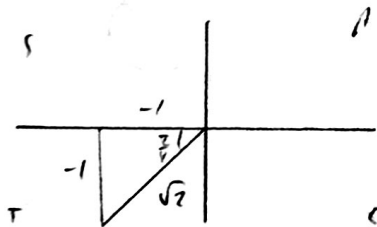
$$= \frac{5\sqrt{2}}{3} + \frac{4\sqrt{2}}{3}$$

$$= \frac{9\sqrt{2}}{3}$$

$$= 3\sqrt{2}$$

(3)

$$\csc \frac{5\pi}{4} = -\sqrt{2}$$



$$\cos \frac{\pi}{6} = \frac{\sqrt{3}}{2}$$

